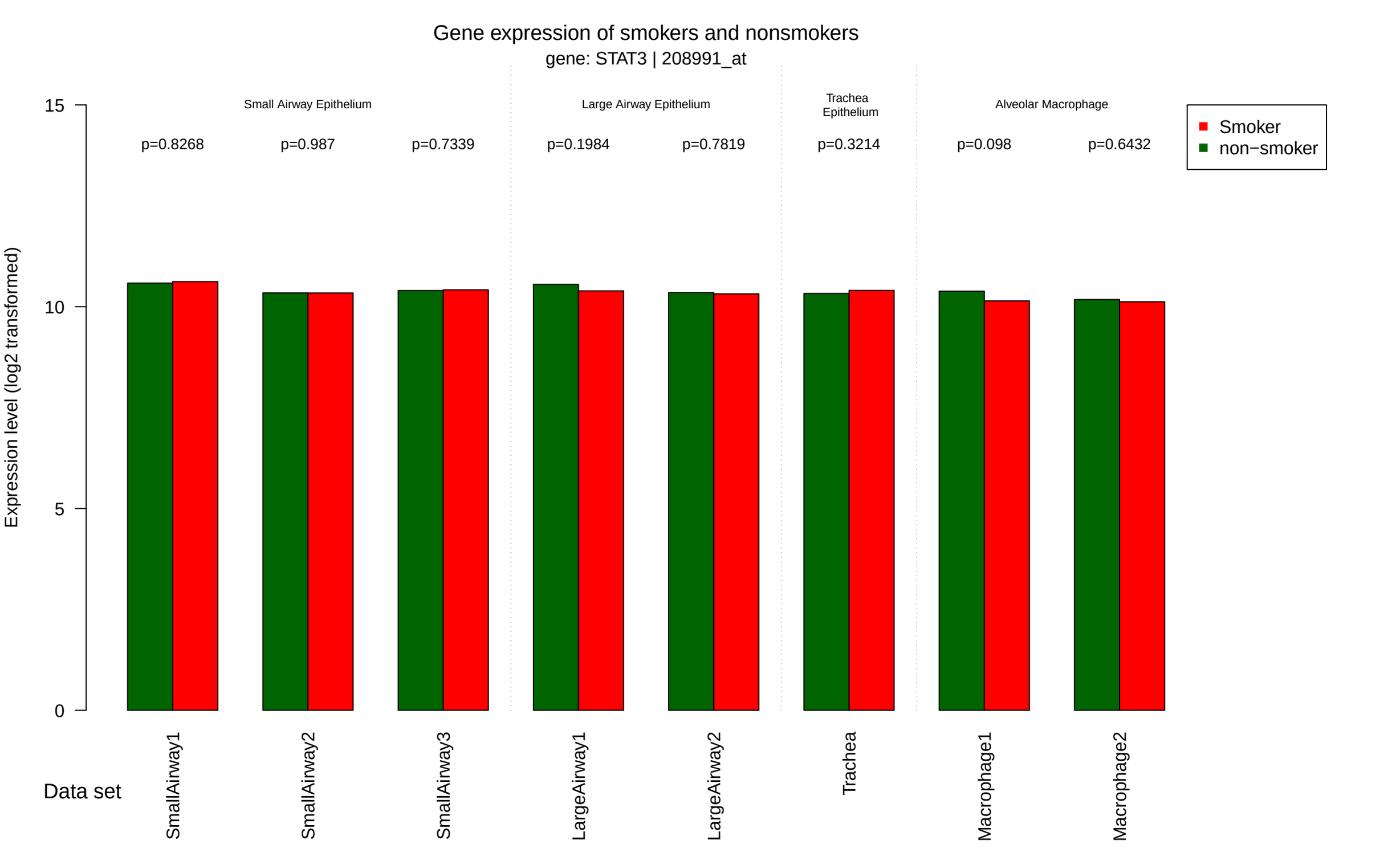
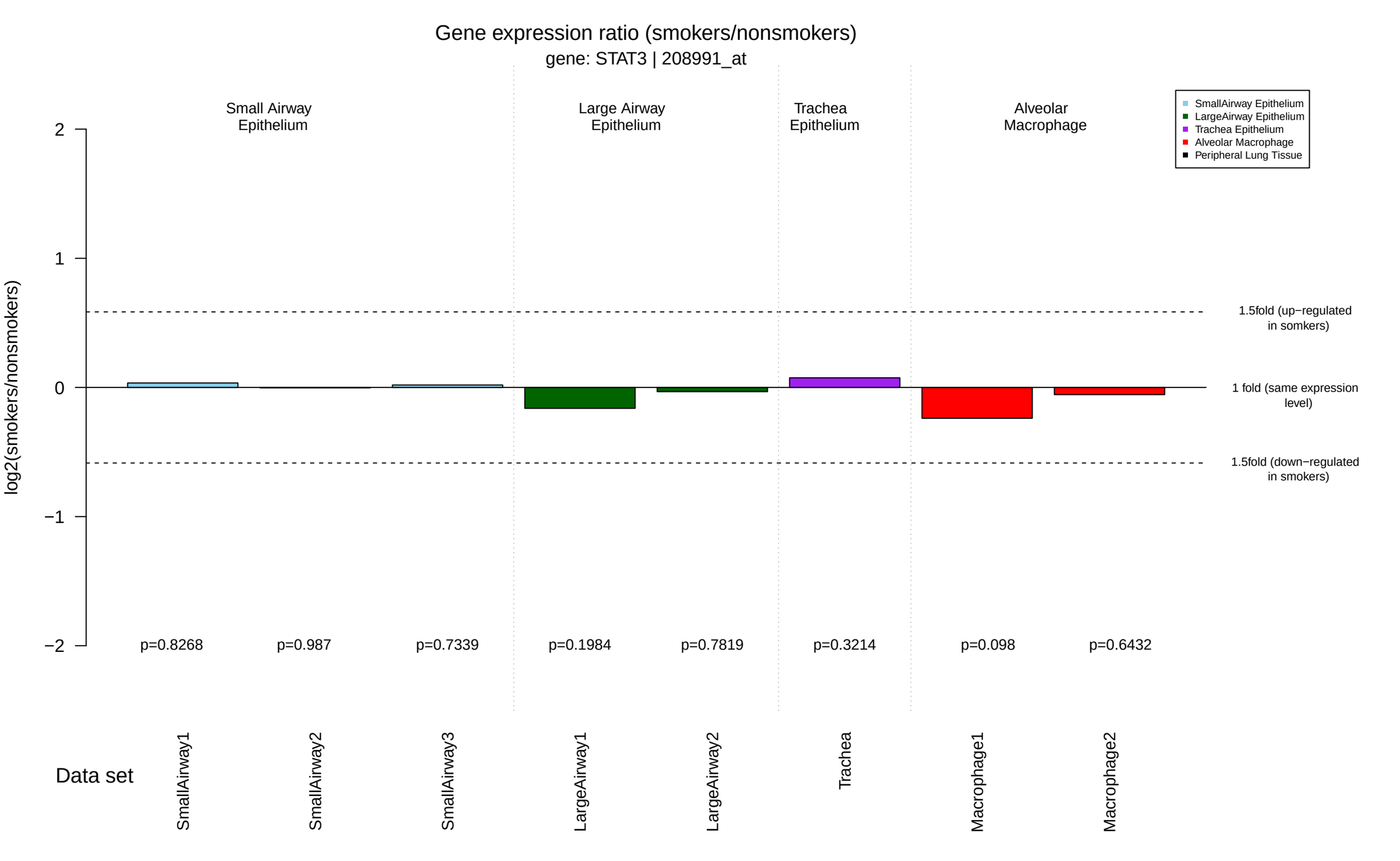
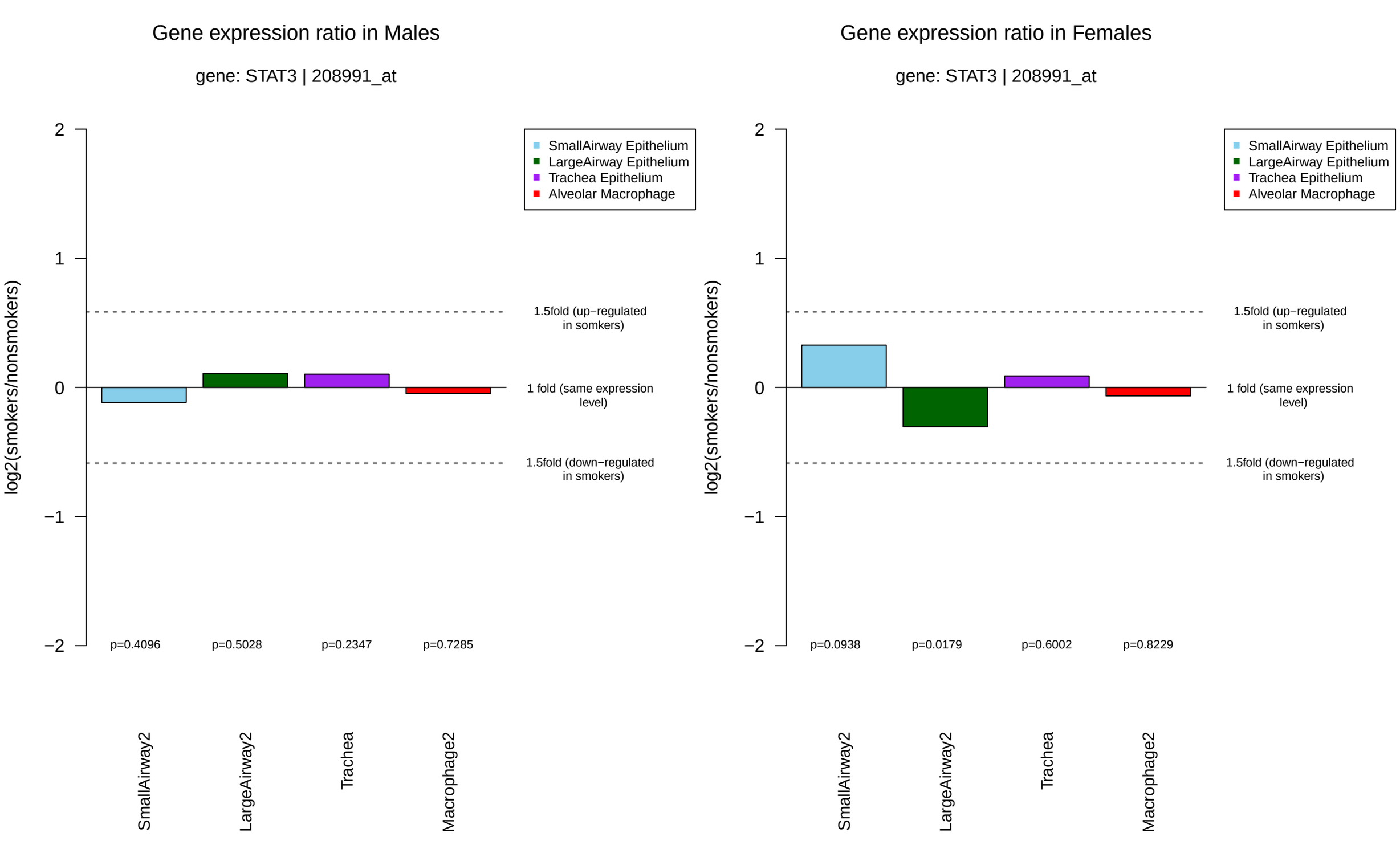
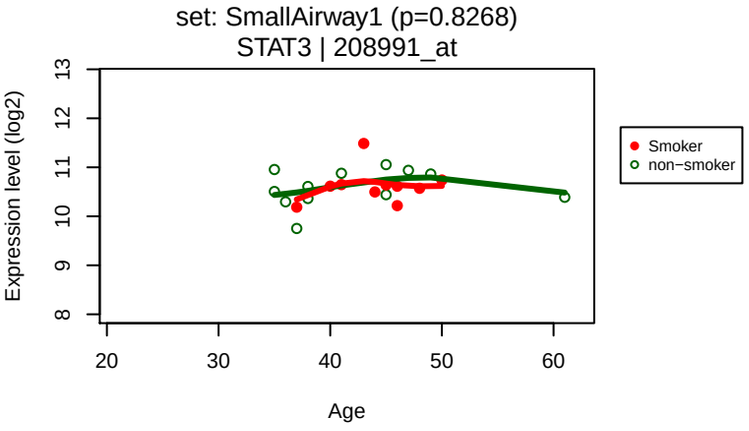
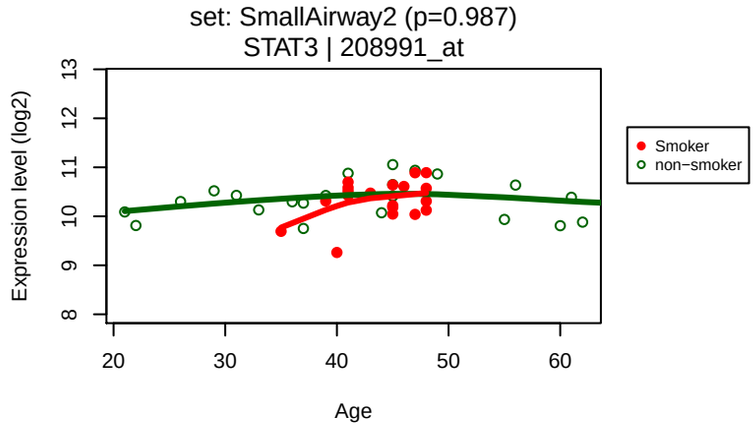
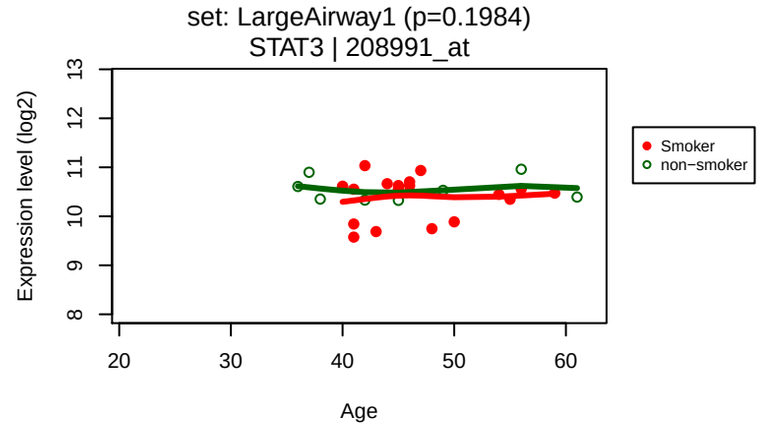

Gene expression of smokers and nonsmokers
gene: STAT3 | 208991_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8268
p=0.987
p=0.7339
p=0.1984
p=0.7819
p=0.3214
p=0.098
p=0.6432
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

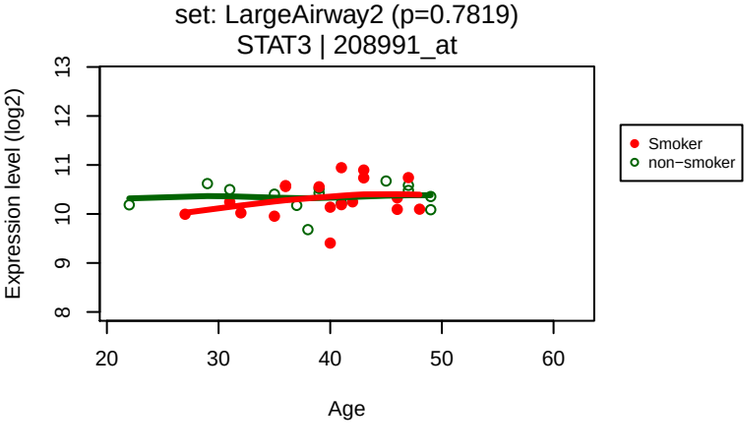

Gene expression ratio (smokers/nonsmokers)
gene: STAT3 | 208991_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8268
p=0.987
p=0.7339
p=0.1984
p=0.7819
p=0.3214
p=0.098
p=0.6432
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

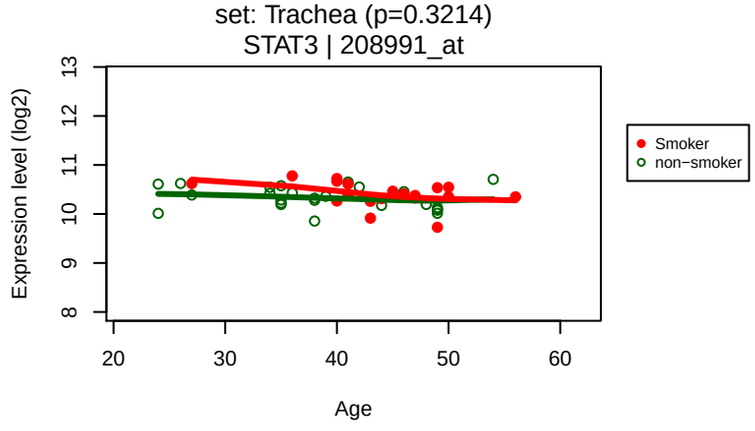

Gene expression ratio in Males
gene: STAT3 | 208991_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4096
p=0.5028
p=0.2347
p=0.7285
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: STAT3 | 208991_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0938
p=0.0179
p=0.6002
p=0.8229
SmallAirway2
LargeAirway2
Trachea
Macrophage2

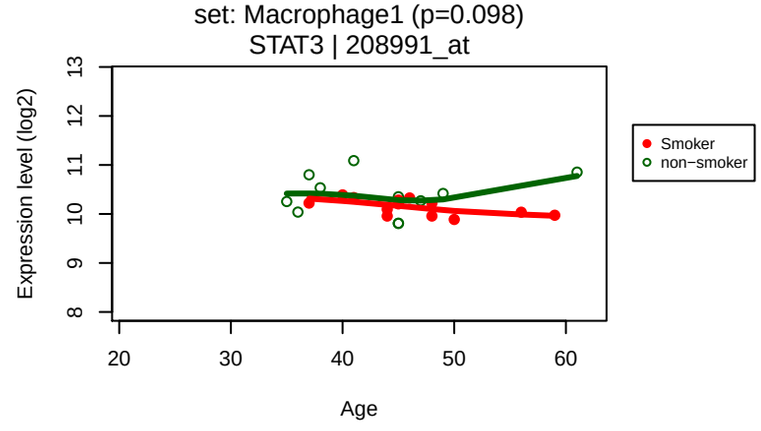

set: SmallAirway1 (p=0.8268)
STAT3 | 208991_at
Expression level (log2)
Age
Smoker
non−smoker

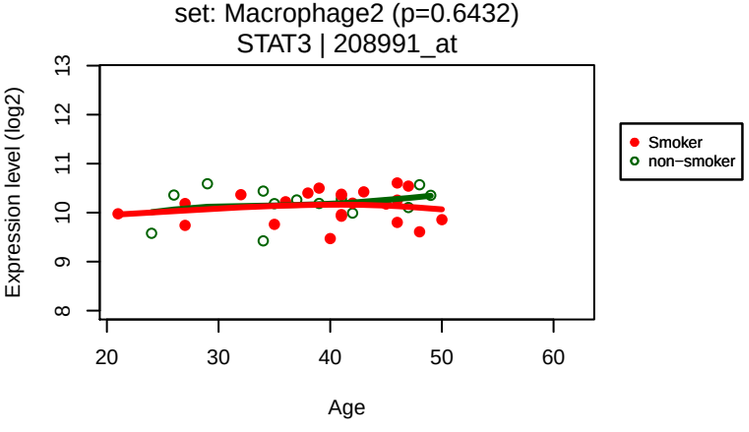

set: SmallAirway2 (p=0.987)
STAT3 | 208991_at
Expression level (log2)
Age
Smoker
non−smoker

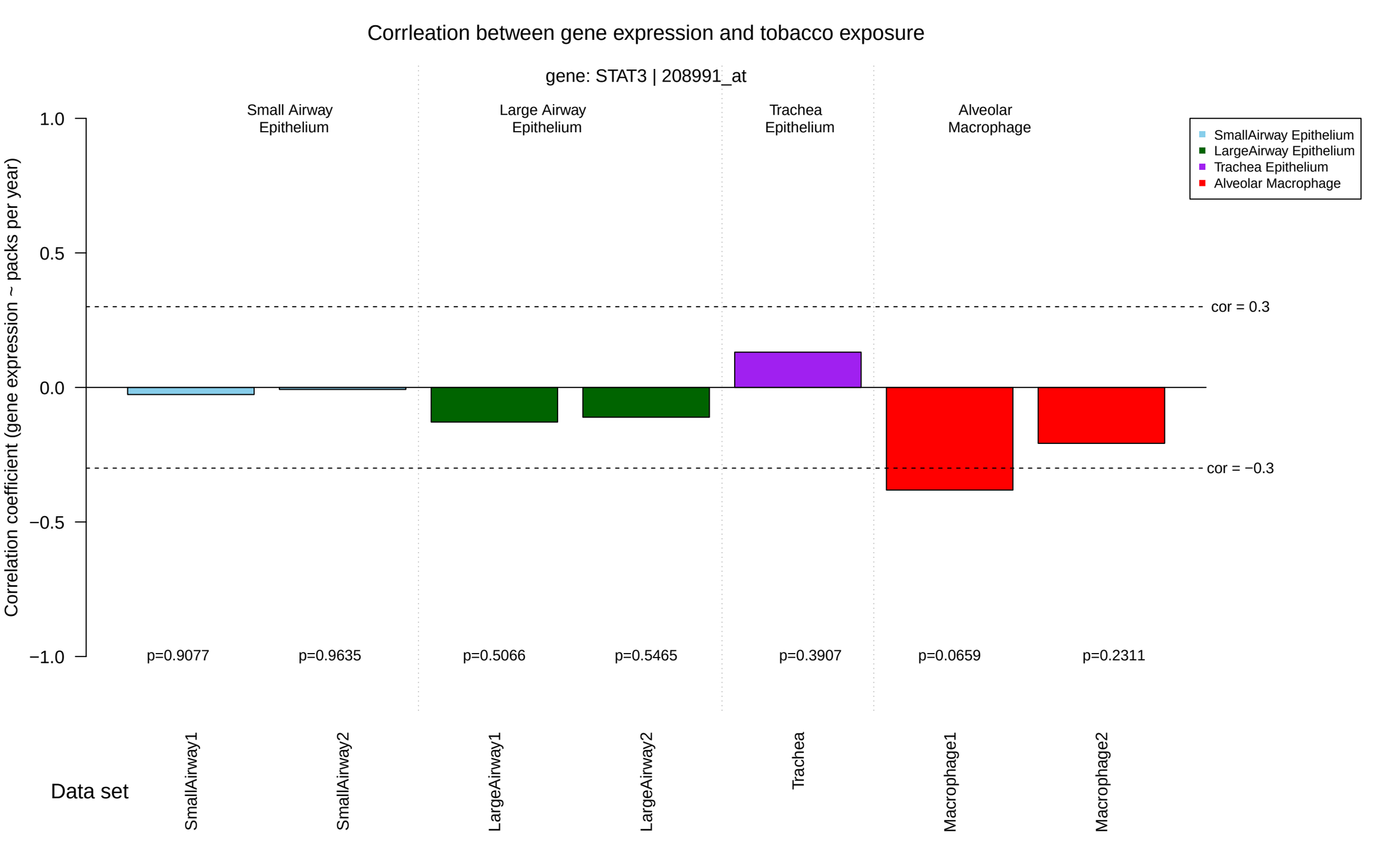

set: LargeAirway1 (p=0.1984)
STAT3 | 208991_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7819)
STAT3 | 208991_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3214)
STAT3 | 208991_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.098)
STAT3 | 208991_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6432)
STAT3 | 208991_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: STAT3 | 208991_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9077
p=0.9635
p=0.5066
p=0.5465
p=0.3907
p=0.0659
p=0.2311
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2